Zackery Pierce

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Education

Wentworth Institute of Technology — Expected August 2028

B.S. in Mechanical Engineering | Available for co-op January–April 2027

Relevant coursework Manufacturing Processes, Design of Machine Elements, Mechanics of Materials, Engineering Statics, Physics I, Physics II, Engineering Graphics, Calculus 3, Differential Equations, Applied Engineering Analysis, Thermodynamics I, Fluid Mechanics, Chemistry I,

Engineering Projects

Wobbler Engine | Manufacturing Lab — September-November 2026

- Machined a 6061 aluminum cylinder plate to drawing specifications on a manual vertical mill. Set work zero with an edge finder and DRO, then faced, drilled, reamed, and hand-tapped ¼-20 UNC and ¼-28 UNF threads.
- Prepared a two-part green-sand mold with sprues, a runner, and a pouring basin.
- Assembled and tested the engine on compressed air, reaching 3,800 rpm.

Threaded Insert for a Gazebo Frame | Personal Project — July 2025

- Measured a failed insert with calipers, modeled the replacement parametrically in SolidWorks, and refined wall thickness and infill across three FDM prints to achieve the threaded connection and press fit.
- Printed the final PLA insert on a Bambu Lab P1S and installed it without modifying the existing assembly.

ThermaVest Heated Jacket Sleeve | Engineering Design | Team of 4 — January–April 2025

- Led CAD on a four-person team and designed a vented SolidWorks enclosure for a 7.4 V, 2,200 mAh Li-ion battery pack, switch, and wiring harness.
- Supported FDM fabrication and assembly. Used FLIR imaging and a multimeter to evaluate sleeve temperature and initial current draw.

Work Experience

Vans | Sales Associate | Boston, MA — November 2025–Present

- Advise customers on product construction, materials, and durability, process purchases, and restock inventory while working 12 hours per week alongside full-time study.

Shaw's | In-Store Shopper | Braintree, MA — July 2022–August 2026

- Fulfilled approximately 20 online grocery orders per shift within delivery windows and used a digital inventory system to verify items and order accuracy.

Skills and Certifications

Machining : Manual vertical mill, ProTRAK Lathe ,DRO, edge finding, workholding, drilling, reaming, and hand tapping

CAD and fabrication : SolidWorks parts, assemblies, and drawings, FDM 3D printing, green-sand mold preparation, welding

Measurement and software : Calipers, FLIR thermal imaging, multimeter, Microsoft Excel, MATLAB and Python (academic familiarity), Stroboscope

Certifications and training : CSWA, OSHA 10-Hour General Industry